Software Implementation and Testing Document

For

Group 28

Version 1.0

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1. Programming Languages (5 points)

The frontend portion of our project is implemented with a react app in typescript. The backend portion of the app is implemented with a .NET project in C#.

2. Platforms, APIs, Databases, and other technologies used (5 points)

We have decided to integrate with Square's platform via Square API as the primary method of accessing restaurant menu and catalog data. Additionally our group created a Supabase project that is connected through the Supabase .NET client. This database will be used to retrieve restaurant data such as ingredients, menu items, item stock, inventory, etc. Currently, in the early stages we are using expo go to deploy the react app and plan to use Vercel for future app deployment. We are utilizing .NET 9 to power the back end portion of the project and have taken advantage of the AutoMapper and Swagger technologies for mapping and testing.

3. Execution-based Functional Testing (10 points)

Interactive Swagger UI

- Used to test the database queries such as get, post, delete
- Will be used down the line to ensure that the C# functions are accurately fetching from and updating our database (hosted by Supabase)

Smoke Tests

- In the Program.cs file of our .NET project
- Supabase smoke test confirms connection to the database at application startup by logging the number of rows retrieved from a table.
- Square API smoke test validates Square catalog integration by printing the catalog items

4. Execution-based Non-Functional Testing (10 points)

Reliability / Error Handling

- Tests are conducted at application startup to verify inputs
- Connection confirmation to database and APIs to avoid crashes

Security

- Confidential values such as API keys, access tokens, etc stored as dotnet user-secrets to avoid hard-coding sensitive information

5. Non-Execution-based Testing (10 points)

All code is peer reviewed by other group members via github pull requests. Feedback is provided to each group member upon creation of a pull request and once the group unanimously approves the new changes, the code is pushed to the Increment 1 branch.